Prodigy Anesthesia Quick Review

Congestive Heart Failure

Is congestive heart failure caused by systolic dysfunction, diastolic dysfunction, or both?	Congestive heart failure is usually associated with systolic dysfunction, which occurs when the heart is unable to pump an adequate amount of blood to meet the metabolic requirements of the body. But, symptoms of heart failure can also occur with diastolic dysfunction as a result of atrial hypertension. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 365-366.*
The left ventricle is most commonly involved in systolic heart failure, and often the right ventricle is involved secondarily, but in what instances is systolic heart failure associated with isolated right ventricular failure?	Advanced disease of the lung parenchyma or pulmonary vasculature *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 365.*
What are the most common causes of left ventricular failure?	Primary myocardial dysfunction associated with coronary artery disease is the most common culprit, but valvular disease, arrythmias, and pericardial disease are also causes of left ventricular failure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 205.*
How does ventricular dilation help compensate for chronic heart failure?	As heart failure ensues, the ejection fraction decreases. As the left ventricle dilates, it accommodates more volume. Thus, the same ejection fraction of an increased volume will still be a normal stroke volume. As venous congestion and ventricular dilatation continues, however, clinical deterioration will eventually occur. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*
How are the mixed venous oxygen tension and arteriovenous oxygen content difference affected by heart failure?	The mixed venous oxygen tension is decreased and the arteriovenous oxygen content difference is increased. In compensated heart failure, the arteriovenous oxygen content difference may be normal, but increases rapidly with any form of cardiac stress. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 366.*
What are the major compensatory mechanisms in the presence of heart failure?	Increased preload and sympathetic tone, activation of the renin-angiotensin system, and ventricular hypertrophy are the compensatory mechanisms that occur in the presence of heart failure. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367-368.*

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How does right ventricular failure lead to ascites?	As the right ventricle fails, venous congestion leads to systemic venous hypertension, which results in peripheral edema, hepatic congestion, and ascites. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*
In patients with chronic congestive heart failure, ventricular dilation occurs as a compensatory mechanism to allow for ejection of sufficient stroke volume. The relationship that predicts the increase in ventricular wall stress as this dilation occurs is attributed to whose law?	LaPlace's relationship states that wall tension is proportional to the product of intraventricular pressure and ventricular wall radius. *Shubert, D & Leyba, J. Chemistry and Physics for Nurse Anesthesia: A Student Centered Approach. 2nd Ed. New York: Springer Publishing Company; 2013: 183-186.*
What is the 'triple therapy' that is the mainstay of pharmacologic treatment of congestive heart failure?	The triple therapy drug regimen used for patients with congestive heart failure consists of an ACE inhibitor, a beta-blocker, and a diuretic (which is often an aldosterone antagonist). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 205.*
Heart failure is most commonly associated with a decrease in cardiac output, but in what instance is it associated with an increased cardiac output?	Sepsis, or any other hypermetabolic state associated with a severe decrease in the systemic vascular resistance can result in high cardiac output heart failure. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*
Are norepinephrine levels increased or decreased in patients with congestive heart failure?	In congestive heart failure, sympathetic activation is increased, which results in increased secretion of norepinephrine. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*
How are vasopressin levels affected by congestive heart failure?	Circulating vasopressin levels are nearly twice the normal value in patients with congestive heart failure. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367-368.*
How do circulating catecholamine levels correlate with the severity of congestive heart failure?	Circulating catecholamine levels increase in direct proportion to the severity of left ventricular dysfunction. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*

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Congestive Heart Failure

How are catecholamine receptors affected by congestive heart failure?	Because of the chronically increased levels of circulating catecholamines associated with congestive heart failure, the response of adrenergic receptors is diminished (down-regulated). *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*
What electrolyte abnormalities are often present in patients with congestive heart failure?	Patients with congestive heart failure often exhibit hyponatremia due to activation of the vasopressin system. Treatment with diuretics to reduce vascular fluid volume may also lead to hypokalemia and hypomagnesemia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367-368.*
How does interstitial edema occur as a result of congestive heart failure?	The chronic increase in circulating catecholamines produces widespread arteriolar vasoconstriction which shunts blood away from the skin, gastrointestinal tract, heart, and brain. As blood flow to the kidneys decreases, the renin-angiotensin-aldosterone axis is activated which results in sodium retention and interstitial edema. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 367.*

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